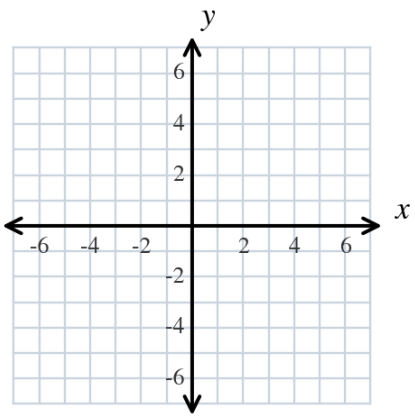
SC Honors Geometry - Nappier
Chapter 4 Review • Transformations

Name: _____



Translations. Find each image or rule.

1. Find the image of A(−6, 3) after a translation along ⟨5, −8⟩.

2. Find the component form of the vector that translates B(4, −2) to B′(−3, 6).

3. Use the rule (x, y) → (x + 2, y − 9). What is the image of C(−1, 4)?

4. Use the same rule. What is the preimage of D′(6, −2)?

5. A figure is translated along ⟨7, −4⟩ and then along ⟨−9, 6⟩. Write ONE rule.

Use the coordinate plane at the right.

6. Graph △EFG with E(−5, 2), F(−2, 4), and G(−1, 0) and its image after the translation (x, y) → (x + 6, y − 5). List the image coordinates.

Reflections. Find each image.

7. H(2, −7) in the x-axis

8. J(−5, 1) in the y-axis

9. K(3, 8) in the line y = x

10. L(−4, 6) in the line y = −x

Reflections in horizontal and vertical lines.

11. Find the image of M(7, −3) after a reflection in the line x = 2.

12. Find the image of N(−1, 5) after a reflection in the line y = −1.

Rotations about the origin. Find each image.

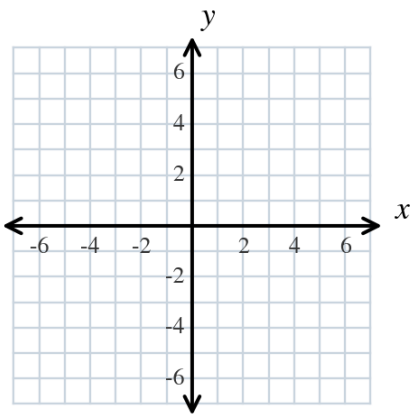
13. P(−2, 6) after a 90° rotation

14. Q(4, −5) after a 180° rotation

15. R(−3, −1) after a 270° rotation

Use the second coordinate plane at the right.

16. Graph △STU with S(1, 3), T(4, 5), and U(5, 1) and its image after a 90° rotation about the origin. List the image coordinates.



Compositions and symmetry.

17. Find the final image of $V\overline{W}$ with V(3, −4) and W(6, −1) after a reflection in the y-axis followed by a 180° rotation about the origin.

18. Find the final image of $X\overline{Y}$ with X(−5, 2) and Y(−2, 4) after the translation (x, y) → (x + 4, y − 7) followed by a 90° rotation about the origin.

19. How many lines of symmetry does a regular hexagon have? Describe every rotation of 180° or less that maps it onto itself.

20. Explain why the image of any figure under a translation, a reflection, or a rotation is always congruent to the preimage.

Congruence transformations.

21. Name the single rigid motion that maps A(−3, 5) to A′(3, 5).

22. Lines m: x = 2 and n: x = 6 are parallel. Reflect in m, then in n. Write ONE rule for the composition, and check it with (1, 3).

23. Reflect a figure in the x-axis, then in the y-axis. What single transformation is this? Justify with (4, −3).

Dilations. All dilations are centered at the origin unless a center is given.

24. Find the image of P(4, −6) after a dilation with k = 3.

25. Find the image of Q(−10, 8) after a dilation with k = 1/2.

26. A dilation maps R(2, −3) to R′(−8, 12). Find the scale factor.

27. Dilate S(5, 1) about the center C(1, −1) with k = 2.

Similarity transformations.

28. Name the similarity transformation that maps △ABC with A(2, 4), B(6, 2), C(4, 8) onto △A′B′C′ with A′(1, 2), B′(3, 1), C′(2, 4).

29. Find the final image of T(3, −2) after a dilation with k = 2, then the translation $(x, y) \rightarrow (x - 5, y + 1)$.

30. △DEF ~ △GHJ with a scale factor of 3. If DE = 4.5 and m∠E = 72°, find GH and m∠H.